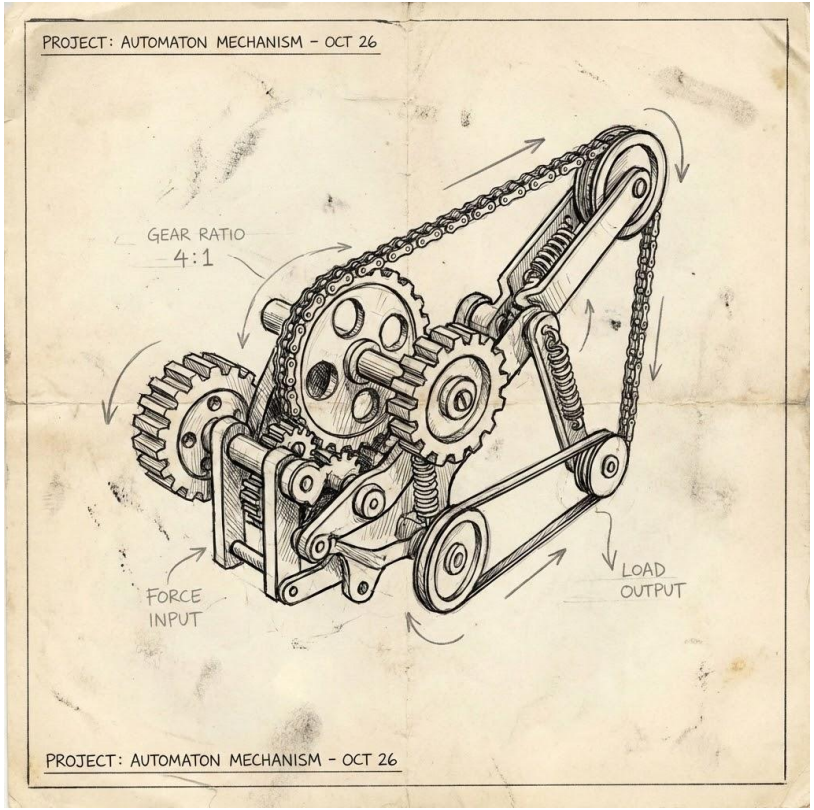
CSC3041F

Control

Krupa Prag
krupa.prag@uct.ac.za



Image: Lucie de Moyencourt



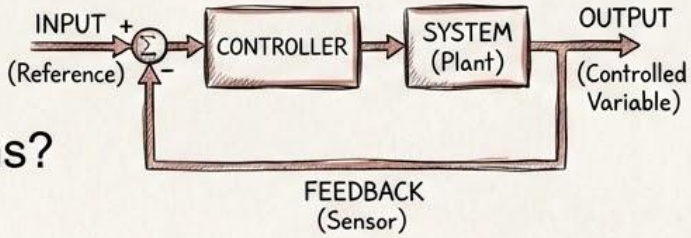
Control Systems

Control System

- A control system manages, commands, or regulates the behaviour of devices or systems using control loops to achieve a desired output.

- What are the components of a control system?

- What are some classifications of control systems?



Open and Closed Loop Control Systems





Open loop control systems

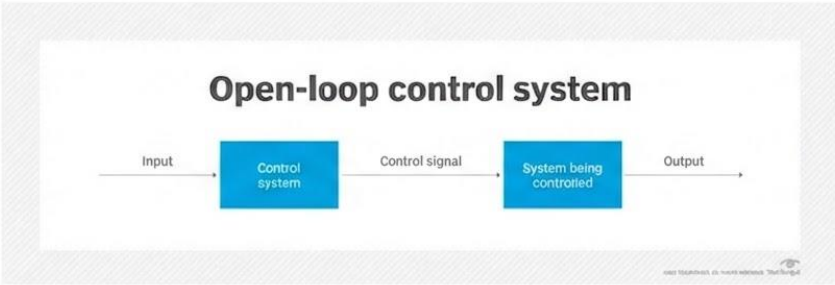
No feedback, fixed operation based on input





Closed loop control systems

Uses feedback to adjust output and maintain desired state


Open-loop control system
Input
Control system
Control signal
System being controlled
Output

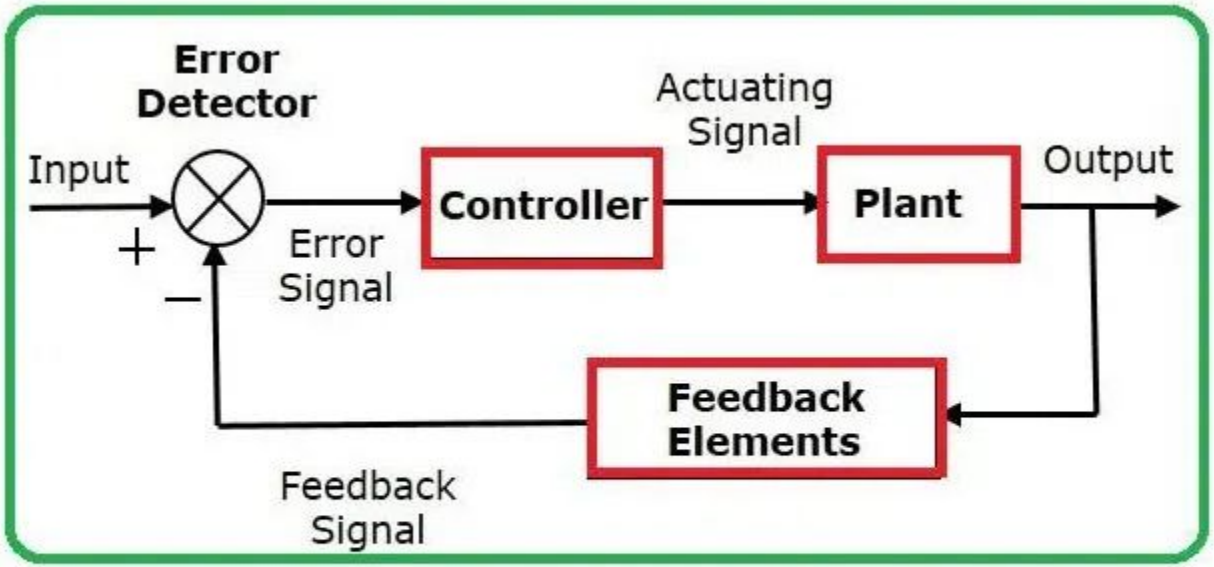
Open-loop Control System

Advantages:

- Simplicity:
  - Very easy to design and build.
- Cost-Effective:
  - Fewer components (no sensors needed).
- *Stability*:
  - Because there is no feedback, the system cannot "over-correct" itself and become unstable or start oscillating.

Disadvantages

- Inaccuracy:
  - It cannot account for changes in the environment (e.g., if the toaster is already hot from a previous round, the second slice will be darker even on the same setting).
- No Disturbance Rejection:
  - It cannot handle external "noise." If a gust of wind cools down a timed heater, an open-loop system won't increase the power to compensate.
- Requires Calibration:
  - You have to know exactly how the system reacts to every input beforehand.



Components of Control Systems

Closed-loop control system

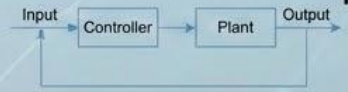
Open-loop Control System

Advantages:

- **Simplicity:** Very easy to design and build.
- **Cost-Effective:** Fewer components (no sensors needed).
- **Stability:** Because there is no feedback, the system cannot 'over-correct' itself and become unstable or start oscillating.

Disadvantages:

- **Inaccuracy:** It cannot account for changes in the environment (e.g., if the toaster is already hot from a previous round, the second slice will be darker even on the same setting).
- **No Disturbance Rejection:** It cannot handle external 'noise.' If a gust of wind cools down a timed heater, an open-loop system won't increase the power to compensate.
- **Requires Calibration:** You have to know exactly how the system reacts to every input beforehand.



State-space



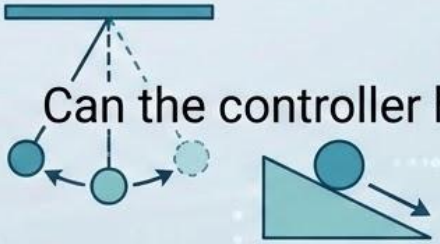
How do we represent the state space?



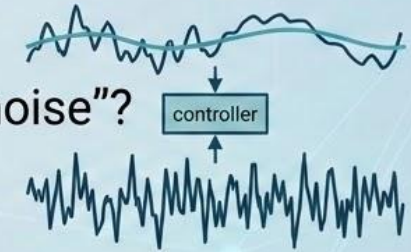
How do we represent the dynamics?

Stability & Objective

- If I nudge the system, does it return to equilibrium or explode?



- Can the controller handle “noise”?





- How do you define the cost function or objective function?

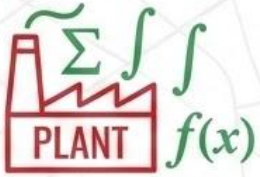
Model-Based Control

- is a type of control system that uses a mathematical "map" of the process it is controlling to make decisions.





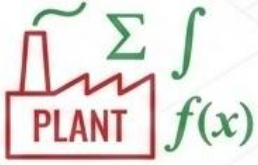
- a model-based controller predicts how the system will behave in the future based on its current state and potential actions.



- The control system is usually the mathematical representation of the Plant.







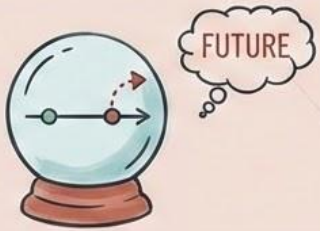
Why use a model?

Mathematical model.

$$\sum \int f(x)$$



Prediction?



Efficiency?



Constraints?



Pros and Cons of Model-Based Control

- When do model-based control work?

When they work:	When they do not work:
✓ When do model-based control work?	✗ When do model-based control not work?

Model-free control

Key Characteristics: Motivation. No mathematical representation.

CONSIDERATION 1

? What are the pros and cons of this?

CONSIDERATION 2

? What do the design of the controllers look like now?

PID Controllers

- "workhorse" of the industry
- PID stands for Proportional, Integral, and Derivative.
- It is a Model-Free closed-loop feedback mechanism.
- What is the objective of the controller?

PID Controller Components

Proportional (P) - "The Present": Adjusts the output based on the current size of the error.



THE PRESENT



THE PAST



THE FUTURE

Integral (I) - "The Past": Adjusts based on the accumulation of past errors to eliminate steady-state offset.

Derivative (D) - "The Future": Adjusts based on the rate of change of the error to predict and dampen future motion.

The PID Equation

Define the components of a PID Controller